

Numerical Integration: A Concise Review Outline

with an appendix on high-dimensional quadrature and volume estimation

Public study version. Personal project details intentionally omitted.

Purpose

These notes organize the standard numerical integration chapter around one core problem:

$$I(f) = \int_a^b f(x) \, dx$$

and its finite approximation by weighted function values. The appendix explains how the same ideas change in high-dimensional volume estimation, where the main obstruction is the *curse of dimensionality*.

1 Corrections to the Interview Summary

Your original outline was mathematically sound in its main direction. The following points should be corrected or sharpened.

1. The trapezoidal rule and midpoint rule are better described as consequences of integrating low-degree local approximations to the integrand, or as elementary interpolatory/rectangle-type formulas. The mean value theorem for integrals motivates the idea but is not the full derivation.
2. Use one consistent convention for the number of Gaussian nodes. With m nodes, Gaussian quadrature has algebraic degree of precision $2m - 1$. If the nodes are indexed as $x_0, \dots, x_n$, there are $n + 1$ nodes and the maximal degree is $2n + 1$.
3. Chebyshev nodes do not simply mean “Gaussian quadrature” in every context. Chebyshev points lead naturally to Clenshaw-Curtis quadrature; Chebyshev-Gauss quadrature is a Gaussian formula for a weighted integral such as $\int_{-1}^1 f(x)(1 - x^2)^{-1/2} \, dx$.
4. Multidimensional integration is not merely a routine extension. Tensor-product rules are conceptually simple but their cost grows exponentially with dimension. This is the curse of dimensionality, and it is central in volume estimation problems.

Polished version of your English overview

Thank you again for the opportunity to interview for the project. I would also like to clarify my understanding of the numerical integration chapter, since I may not have expressed some mathematical ideas clearly enough in English during the interview.

The basic problem in numerical integration is to approximate a definite integral by a finite weighted sum of function values. Starting from elementary rules such as the midpoint rule and the trapezoidal rule, one naturally arrives at the general form

$$\int_a^b f(x) \, dx \approx \sum_{i=0}^n A_i f(x_i),$$

where the nodes x_i and weights A_i determine the accuracy and stability of the formula. Interpolatory quadrature is obtained by replacing the integrand by its interpolation polynomial and integrating the interpolant exactly. Newton-Cotes formulas correspond to equally spaced nodes, with the trapezoidal rule and Simpson's rule as basic examples. For long intervals, composite quadrature formulas are obtained by subdividing the interval and applying a local rule on each subinterval.

To measure the quality of a quadrature formula, we use algebraic degree of precision and error terms. A formula has algebraic degree of precision r if it is exact for all polynomials of degree at most r and fails, in general, for some polynomial of degree $r + 1$. Romberg integration improves the composite trapezoidal rule through Richardson extrapolation, eliminating leading error terms and increasing the order of accuracy.

A more advanced idea is to choose the nodes optimally. With m nodes, Gaussian quadrature achieves the maximal algebraic degree of precision $2m - 1$ by choosing the nodes as the zeros of suitable orthogonal polynomials. Legendre nodes are used for unweighted integrals on a finite interval, while other choices such as Chebyshev, Laguerre, and Hermite nodes are adapted to different weights or domains. In high-dimensional problems, tensor-product quadrature is the direct extension, but it suffers from exponential growth in the number of points. Therefore, Monte Carlo methods, quasi-Monte Carlo methods, sparse grids, symmetry reduction, and problem-specific transformations become important tools for volume estimation and related computational geometry problems.

2 Basic Problem and General Quadrature Formula

A numerical quadrature formula approximates a linear functional:

$$I(f) = \int_a^b f(x) \, dx$$

by

$$Q_n(f) = \sum_{i=0}^n A_i f(x_i), \quad a \leq x_0 < \cdots < x_n \leq b.$$

Here x_i are *nodes* and A_i are *weights*. The error functional is

$$E_n(f) = I(f) - Q_n(f).$$

The whole chapter can be read as a study of how to choose nodes and weights to make $E_n(f)$ small for useful classes of functions.

3 Elementary Rules

Let $h = b - a$.

3.1 Midpoint rule

$$M(f) = hf\left(\frac{a+b}{2}\right), \quad I(f) - M(f) = \frac{h^3}{24}f''(\xi), \quad \xi \in (a, b).$$

3.2 Trapezoidal rule

$$T(f) = \frac{h}{2}(f(a) + f(b)), \quad I(f) - T(f) = -\frac{h^3}{12}f''(\xi), \quad \xi \in (a, b).$$

3.3 Simpson's rule

$$S(f) = \frac{h}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right],$$

$$I(f) - S(f) = -\frac{h^5}{2880}f^{(4)}(\xi), \quad \xi \in (a, b).$$

Simpson's rule is exact for all polynomials of degree at most 3.

4 Interpolatory Quadrature

Choose distinct nodes $x_0, \dots, x_n$ and let $\ell_i(x)$ be the Lagrange basis polynomial:

$$\ell_i(x) = \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}.$$

The interpolation polynomial is

$$p_n(x) = \sum_{i=0}^n f(x_i)\ell_i(x).$$

The interpolatory quadrature formula is obtained by integrating p_n :

$$Q_n(f) = \int_a^b p_n(x) \, dx = \sum_{i=0}^n A_i f(x_i), \quad A_i = \int_a^b \ell_i(x) \, dx.$$

Thus the weights are determined solely by the nodes and the interval, not by f .

5 Algebraic Degree of Precision

A quadrature formula Q has algebraic degree of precision r if

$$Q(p) = I(p) \quad \text{for all } p \in \mathbb{P}_r,$$

but the equality does not hold for all $p \in \mathbb{P}_{r+1}$.

Equivalently, it is enough to check monomials:

$$\sum_i A_i x_i^k = \int_a^b x^k \, dx, \quad k = 0, 1, \dots, r.$$

For an interpolatory formula with $n+1$ nodes, the degree is at least n . Symmetry may raise it, as in Simpson's rule.

6 Newton-Cotes Formulas

Newton-Cotes formulas are interpolatory quadrature formulas with equally spaced nodes:

$$x_i = a + ih, \quad qquad h = \frac{b-a}{n}, \quad i = 0, \dots, n.$$

They have the form

$$\int_a^b f(x) dx \approx h \sum_{i=0}^n \alpha_i f(x_i).$$

Examples:

Rule	Formula	Degree of precision
Trapezoidal	$\frac{h}{2}(f_0 + f_1)$	1
Simpson	$\frac{h}{3}(f_0 + 4f_1 + f_2)$	3
3/8 rule	$\frac{3h}{8}(f_0 + 3f_1 + 3f_2 + f_3)$	3

For high-degree closed Newton-Cotes rules, some weights become negative, which may lead to cancellation and numerical instability. This is one reason why high-order rules based on better node distributions are preferred.

7 Composite Quadrature Formulas

For a long interval, subdivide

$$a = x_0 < x_1 < \dots < x_N = b, \quad H = \frac{b-a}{N}.$$

7.1 Composite trapezoidal rule

$$T_N(f) = H \left[\frac{1}{2}f(x_0) + \sum_{j=1}^{N-1} f(x_j) + \frac{1}{2}f(x_N) \right].$$

If $f \in C^2[a, b]$,

$$I(f) - T_N(f) = -\frac{b-a}{12} H^2 f''(\xi).$$

7.2 Composite midpoint rule

$$M_N(f) = H \sum_{j=0}^{N-1} f\left(a + \left(j + \frac{1}{2}\right) H\right),$$

$$I(f) - M_N(f) = \frac{b-a}{24} H^2 f''(\xi).$$

7.3 Composite Simpson rule

For N even,

$$S_N(f) = \frac{H}{3} \left[f(x_0) + f(x_N) + 4 \sum_{j \text{ odd}} f(x_j) + 2 \sum_{j \text{ even}, 0 < j < N} f(x_j) \right],$$

$$I(f) - S_N(f) = -\frac{b-a}{180} H^4 f^{(4)}(\xi).$$

8 Richardson Extrapolation and Romberg Integration

Suppose an approximation has an asymptotic expansion

$$A(h) = I + c_1 h^p + c_2 h^{p+q} + \dots$$

Then

$$\frac{2^p A(h/2) - A(h)}{2^p - 1} = I + O(h^{p+q}).$$

This is Richardson extrapolation.

For the composite trapezoidal rule,

$$T(h) = I + c_1 h^2 + c_2 h^4 + c_3 h^6 + \dots$$

Romberg integration builds a triangular table:

$$R_{k,0} = T(h_k), \quad h_k = \frac{b-a}{2^k},$$

$$R_{k,j} = R_{k,j-1} + \frac{R_{k,j-1} - R_{k-1,j-1}}{4^j - 1}, \quad j = 1, \dots, k.$$

Each extrapolation step eliminates the next even-power error term.

9 Gaussian Quadrature

For a weighted integral

$$I(f) = \int_a^b w(x) f(x) dx, \quad w(x) > 0,$$

Gaussian quadrature with m nodes has the form

$$Q_m(f) = \sum_{i=1}^m \lambda_i f(x_i).$$

The nodes x_i are chosen as the zeros of the degree- m orthogonal polynomial with respect to

$$\langle p, q \rangle = \int_a^b w(x) p(x) q(x) dx.$$

The formula has maximal algebraic degree of precision

$$2m - 1.$$

Common choices:

Formula	Domain/weight	Orthogonal polynomials
Gauss-Legendre	$[-1, 1], w = 1$	Legendre
Gauss-Chebyshev	$[-1, 1], w = (1 - x^2)^{-1/2}$	Chebyshev
Gauss-Laguerre	$[0, \infty), w = e^{-x}$	Laguerre
Gauss-Hermite	$(-\infty, \infty), w = e^{-x^2}$	Hermite

If f is sufficiently smooth, the error involves a $2m$ -th derivative. In the standard positive-weight setting,

$$I(f) - Q_m(f) = \frac{f^{(2m)}(\xi)}{(2m)!} \int_a^b w(x) \pi_m(x)^2 dx,$$

where $\pi_m(x) = \prod_{i=1}^m (x - x_i)$ and $\xi \in (a, b)$.

10 Adaptive Quadrature

Adaptive quadrature allocates more nodes where the integrand is difficult, such as near sharp gradients, singularities, or oscillations.

A standard strategy is:

1. compute two approximations on an interval, for instance $S([a, b])$ and $S([a, m]) + S([m, b])$;
2. estimate the local error by their difference;
3. subdivide intervals whose estimated error is too large;
4. stop when the global tolerance is met.

For Simpson's rule,

$$\text{err}([a, b]) \approx \frac{S([a, m]) + S([m, b]) - S([a, b])}{15}.$$

Adaptive quadrature is not a new kind of formula; it is a control strategy for distributing computational work.

11 Multidimensional Integration: Basic Extension

For a rectangular domain

$$\Omega = \prod_{j=1}^d [a_j, b_j],$$

the integral is

$$I(f) = \int_{a_1}^{b_1} \cdots \int_{a_d}^{b_d} f(x_1, \dots, x_d) \, dx_d \cdots dx_1.$$

11.1 Tensor-product quadrature

If a one-dimensional rule is

$$Q^{(j)}(g) = \sum_{i=1}^{m_j} w_i^{(j)} g(x_i^{(j)}),$$

then the tensor-product rule is

$$Q_d(f) = \sum_{i_1=1}^{m_1} \cdots \sum_{i_d=1}^{m_d} \left(\prod_{j=1}^d w_{i_j}^{(j)} \right) f(x_{i_1}^{(1)}, \dots, x_{i_d}^{(d)}).$$

If $m_j = m$ for all j , the number of function evaluations is

$$m^d.$$

This is the most direct place where the curse of dimensionality appears.

11.2 Iterated integration

Fubini-type reduction rewrites a multidimensional integral as repeated one-dimensional integrals. Numerically, this is useful when inner integrals have special structure or can be computed analytically. However, if each direction is discretized uniformly, the total cost remains exponential in d .

11.3 General domains

For non-rectangular domains, one usually combines:

- coordinate transformations to a reference region;
- decomposition into simple cells, such as triangles, tetrahedra, or simplices;
- local quadrature rules on each cell;
- symmetry reduction if the body has a nontrivial symmetry group.

For volume estimation, the integral is often

$$\text{Vol}(K) = \int_{\mathbb{R}^d} \mathbf{1}_K(x) \, dx,$$

or, after bounding-box reduction,

$$\text{Vol}(K) = \text{Vol}(B) \mathbb{P}(X \in K), \quad X \sim \text{Unif}(B).$$

A Appendix: Fighting the Curse of Dimensionality in Volume Estimation

A.1 Why tensor products fail

Suppose a one-dimensional quadrature rule uses $m = 20$ nodes. In dimension 4, the tensor-product cost is $20^4 = 160000$ samples. In dimension 6, it is $20^6 = 64,000,000$ samples. Even before evaluating a complicated membership test for a convex body, the grid has become expensive.

A.2 Monte Carlo integration

For a bounded measurable set $K \subset B \subset \mathbb{R}^d$,

$$\text{Vol}(K) = \text{Vol}(B) \mathbb{E}[\mathbf{1}_K(X)], \quad X \sim \text{Unif}(B).$$

With samples $X_1, \dots, X_N$,

$$\widehat{\text{Vol}}(K) = \text{Vol}(B) \frac{1}{N} \sum_{i=1}^N \mathbf{1}_K(X_i).$$

The standard deviation is

$$\text{Vol}(B) \sqrt{\frac{p(1-p)}{N}}, \quad p = \mathbb{P}(X \in K).$$

The key advantage is that the nominal convergence rate $O(N^{-1/2})$ does not explicitly contain d . The disadvantage is slow convergence and random noise.

A.3 Variance reduction

Common variance-reduction ideas include:

- using a tighter bounding body B so that p is not too small;
- exploiting symmetries to reduce the integration domain;
- stratified sampling, where B is decomposed into subregions and sampled more evenly;
- importance sampling, replacing a uniform distribution by a distribution better adapted to the geometry;
- control variates, when the volume of a related body is known.

A.4 Quasi-Monte Carlo

Quasi-Monte Carlo replaces random points by deterministic low-discrepancy sequences. For sufficiently regular integrands, this often improves practical convergence. It is especially useful when the effective dimension is lower than the ambient dimension.

A.5 Sparse grids and Smolyak construction

Sparse grids combine low-level tensor-product rules in a way that avoids using the full grid. The guiding principle is to keep mixed-resolution information and discard excessively fine simultaneous refinement in every coordinate. For smooth functions, sparse grids can be far cheaper than full tensor products. They are less natural for discontinuous indicator functions $\mathbf{1}_K$, but may still be useful after smoothing, transforming variables, or integrating radial/support-function descriptions.

A.6 Convex-body volume estimation: practical checklist

For a volume-estimation problem in dimension 4 or higher:

1. Find a good analytic description of the body: inequalities, support function, radial function, or symmetry-reduced coordinates.
2. Validate the code on dimensions 2 and 3, where theoretical volumes or high-accuracy benchmarks are available.
3. Compare at least two independent methods, for example deterministic quadrature and Monte Carlo.
4. Track both numerical error and modeling error: discretization, sampling error, boundary handling, and membership-test tolerance.
5. Use symmetry aggressively. If the body has tetrahedral, permutation, or reflection symmetries, integrate over a fundamental region and multiply by the group order.
6. Report confidence intervals or convergence tables, not just a single number.

B Source Map

The main one-dimensional integration sequence follows standard numerical-analysis treatments of Newton-Cotes formulas, Peano error representation, extrapolation, and Gaussian integration. The multidimensional appendix follows the standard progression: tensor products, standard-domain integration, and Monte Carlo, with additional modern context on sparse grids and high-dimensional volume estimation.